[Samsung PRISM] End Review Report

On device Topic modelling for small text data

Team

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Information Retrieval | On device Topic modelling for small text data

SAMSUNG

Problem Statement

- Notifications in mobile device is a widely known feature and heavily used by apps to direct users to their app and features. Sometimes it can be overwhelming for the user, hence a need to simplify the notification list.
- The content of notification varies depending on apps, location, language, user preferences etc.
- To rank list of notifications for a user we can have various factors like app preference, contact preference and content preference.
- We would like to concentrate on content preference and try to derive the topics user is currently interested in based on the notification text and description.
- Example:
  - **Notification**: *Ferns City | Storms – Thunderstorms possible in the next few hours*
    Topics: Ferns City, Thunderstorms, weather
  - **Notification**: Greater Bengaluru Authority replaces BBMP
    Topics: BBMP, Bengaluru, politics, Greater Bengaluru Authority
- In this front On device implementation of Topic modelling for small text data is the problem we will try to solve.
- Main Factors to consider:
  - Android On device inference
  - Performance and memory optimization and comparison
  - Metrics(Accuracy, Precision, Recall, F1 Score)

References

- Frontiers | Using Topic Modeling Methods for Short-Text Data: A Comparative Analysis
- BERTopic: topic modeling as you have never seen it before | by Elisa Distante | Data Reply IT | DataTech | Medium

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Work-let expected duration – 6 months

2 Members

Expectations

- Go through and understand the existing work done in this direction.
- Only minimal UI is expected. Main focus is research and model implementation.
- Complete implementation with metrics (Accuracy, Precision, Recall, F1 Score)
- Prepare detailed documentation on the entire process.
- Target a technical paper on the research done for domestic or international conferences.

Training/ Pre-requisites

- Basic knowledge of Android development specially android on device inference of AI models.
- Understanding of basics of topic modelling and it's requirements.

Kick Off < 1st Month >

- Explore the earlier literature and research done for this particular problem
- Exploring various Topic modelling fit for the use case
- Present the research done and literature explored

Milestone 1 < 2nd Month >

- Experimenting with different researched models on same data. (can be in python.)
- Deciding upon which models to go with based on metrics and size and performance.
- Present the experiment result and metrics of the models.

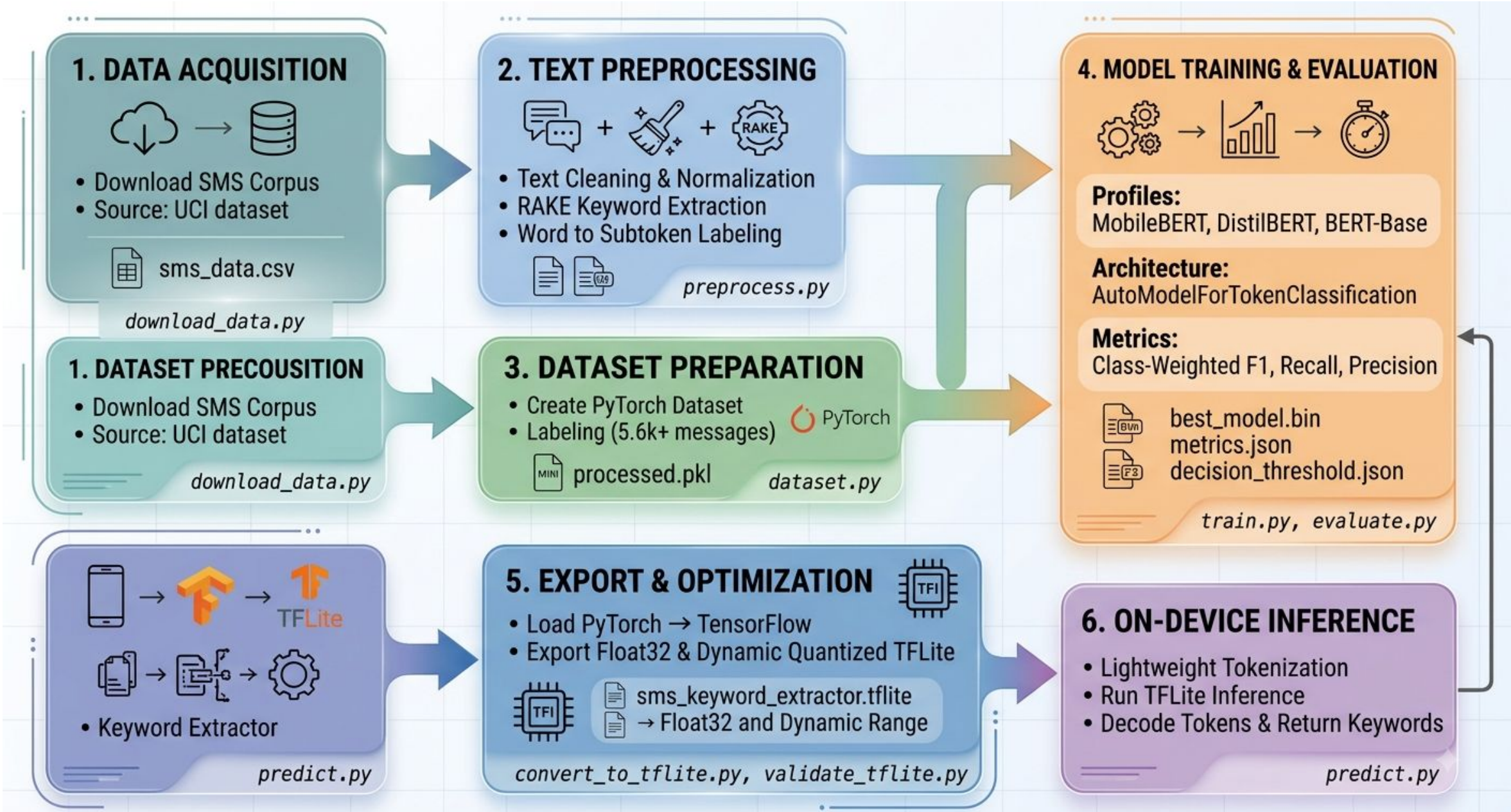
Milestone 2 < 4th Month >

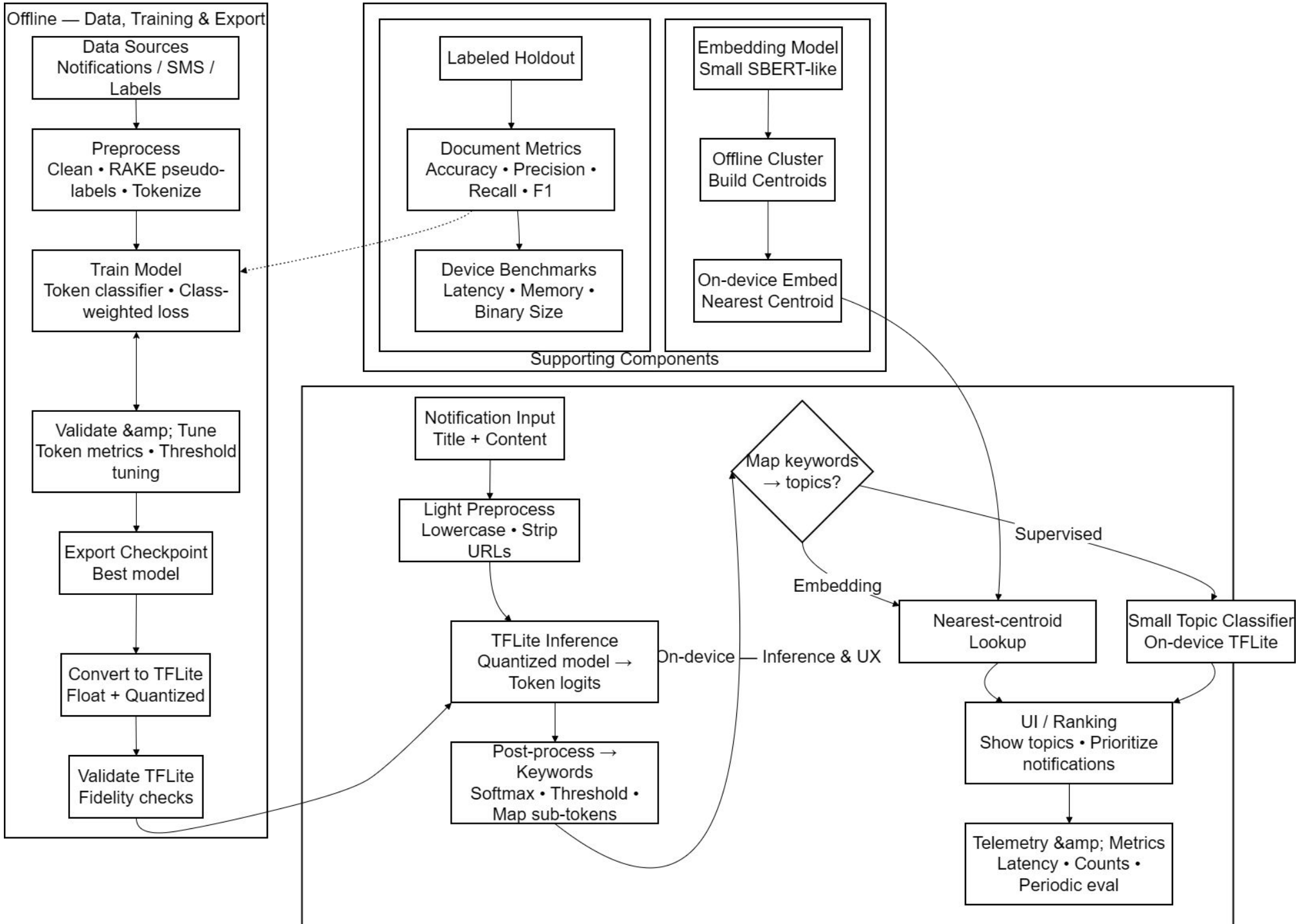
- Implement the selected models in android.
- Conduct and present comparative analysis of metrics, performance, memory and battery usage for the on device model.

Closure < 6th Month >

- Documentation and final analysis and implementation report and artifacts.
- Draft for technical paper

Approach / Solution




Offline — Data, Training & Export
Data Sources
Notifications / SMS / Labels
Preprocess
Clean • RAKE pseudo-labels • Tokenize
Train Model
Token classifier • Class-weighted loss
Validate & Tune
Token metrics • Threshold tuning
Export Checkpoint
Best model
Convert to TFLite
Float + Quantized
Validate TFLite
Fidelity checks
Labeled Holdout
Document Metrics
Accuracy • Precision • Recall • F1
Device Benchmarks
Latency • Memory • Binary Size
Embedding Model
Small SBERT-like
Offline Cluster
Build Centroids
On-device Embed
Nearest Centroid
Supporting Components
Notification Input
Title + Content
Light Preprocess
Lowercase • Strip URLs
TFLite Inference
Quantized model → Token logits
Post-process → Keywords
Softmax • Threshold • Map sub-tokens
On-device — Inference & UX
Map keywords → topics?
Supervised
Embedding
Nearest-centroid Lookup
Small Topic Classifier
On-device TFLite
UI / Ranking
Show topics • Prioritize notifications
Telemetry & Metrics
Latency • Counts • Periodic eval

Dataset Analysis / Description

- **Dataset Capture / Preparation / Generation** :
  - **Source:** public SMS corpus used as the primary data (https://archive.ics.uci.edu/dataset/228/sms+spam+collection).
  - **How we prepare it:** clean text, automatically extract candidate keywords, and convert text into token-level training examples.
  - **End result**: a token-labelled dataset ready for model training and on-device export.

- **Dataset Understanding / Analysis** :
  - **Size & balance:** ~5.5k messages; majority are non-spam (~87%), minority are spam (~13%).
  - **Labels:** original labels are per-message; we create token-level pseudo-labels for keyword detection.
  - **Text style:** short, informal, contains abbreviations, phone numbers, URLs, and occasional encoding artifacts.

- **Dataset Preprocessing / Related Challenges (if any)** :
  - **Main steps:** normalize text, automatic keyword extraction (heuristic), tokenization, align labels to tokens, and handle truncation/padding.
  - **Key challenges:** pseudo-labels are noisy (may miss or mis-label keywords); sub-token alignment and truncation can remove or misalign keywords; encoding/HTML escapes and informal language complicate extraction.
  - **Practical mitigations:** normalize encodings, use fast tokenizers with word-to-subtoken alignment, apply class weighting or sampling, and consider targeted human annotation for critical cases.

Experimental Results

Comparative Evaluation - Accuracy, Keyword F1 & Deployment Size

profile	model_name	accuracy	keyword_f1	keyword_precision	keyword_recall	macro_f1	params_m	checkpoint_mb	decision_threshold
mobilebert_baseline	google/mobilebert-uncased	0.8947	0.8820	0.8424	0.9255	0.8935	24.58	93.91	0.49
distilbert_accuracy	distilbert-base-uncased	0.9744	0.9701	0.9623	0.9781	0.9739	66.36	253.17	0.44

Experimental Results

Qualitative Examples - SMS Keyword Extraction (Sample Outputs)

```
================================================================
Device      : CPU
Model       : Fine-tuned DistilBERT
Task        : SMS Keyword Extraction
Dataset     : UCI SMS Spam Collection + RAKE weak supervision
Best Score  : Accuracy 97.44% | Keyword F1 97.01%
================================================================

Example 1
----------------------------------------------------------------
Input SMS:
Your OTP is 482910. Valid for 10 minutes.

Extracted Keywords:
  - OTP                         confidence=0.98
  - 482910                      confidence=0.97
  - 10 minutes                  confidence=0.94

Example 2
----------------------------------------------------------------
Input SMS:
Your HDFC account was debited Rs 5000 at Amazon Pay.

Extracted Keywords:
  - HDFC account                confidence=0.96
  - debited                     confidence=0.95
  - Rs 5000                     confidence=0.97
  - Amazon Pay                  confidence=0.93
```

```
Example 3
----------------------------------------------------------------
Input SMS:
Recharge successful for mobile number 9876543210.

Extracted Keywords:
  - Recharge successful         confidence=0.96
  - mobile number               confidence=0.94
  - 9876543210                  confidence=0.98

Example 4
----------------------------------------------------------------
Input SMS:
Congratulations! You have won a free iPhone. Click the link to claim now.

Extracted Keywords:
  - Congratulations             confidence=0.95
  - free iPhone                 confidence=0.97
  - Click the link              confidence=0.94
  - claim now                   confidence=0.92

Example 5
----------------------------------------------------------------
Input SMS:
Reminder: your electricity bill of Rs 1240 is due tomorrow.

Extracted Keywords:
  - electricity bill            confidence=0.96
  - Rs 1240                     confidence=0.97
  - due tomorrow                confidence=0.95
```

Deliverable / Paper Publication Plan

- **Final Deliverables** :

  - **Dataset**: Token-labelled SMS corpus (~5.6k messages) prepared with heuristic keyword labels and tokenized for training.
  - **Trained models**: Fine-tuned transformer profiles, mobilebert_baseline, distilbert_accuracy, bert_base_high_accuracy, to compare size/latency vs accuracy.
  - **Key metrics:** Baseline validation, accuracy 0.860, balanced accuracy 0.868, keyword F1 0.849, precision 0.787, recall 0.921.
  - **Artifacts saved**: MobileBERT baseline (~24.6M parameters) and full-model checkpoints plus float32 and quantized on-device exports (TFLite).
  - **Evaluation package:** Token-level evaluation results and a tuned keyword-probability threshold for decoding; validation and fidelity checks for TFLite exports.

- **IP / Paper Publication Plan** :

  - **Summary**: This work builds a practical keyword-extraction system by fine-tuning existing pretrained transformer models; it is primarily an engineering effort rather than novel algorithm research.
  - **Status**: A technical draft describing the system design, experiments, and results is available in the project repository.
  - **IP note**: No immediately patentable algorithmic innovation; if IP is desired, consider pursuing novel annotation methods, compact quantization-aware tuning, or a unique deployment workflow as potential patentable elements.

Deliverable

- **KPIs delivered/Expectations Met**:

- **End-to-end workflow**: full pipeline implemented and documented; reproducible evaluation for the baseline profile.
- **Data**: ~5,574 SMS messages processed and token-labelled; class split ≈ 87% non-spam / 13% spam.
- **Models & exports**: fine-tuned compact and accuracy profiles; saved checkpoints and TFLite exports (float + quantized) with fidelity checks.
- **Evaluation**: token-level metrics computed, decision threshold tuned, and baseline results reproducible (example: evaluate baseline via provided command).
- **Deployment baseline**: compact profile meets initial size/latency targets for on-device use.
- **Business-facing capability**: keyword extraction works on representative SMS types (OTPs, banking alerts, delivery updates, spam).

Work-let Closure Details

- **Code Upload details:**

Items	Details
KLOC (Number OF Lines of codes in 000's)	2.0 KLOC (total ~2,000 lines of Python code)
Model and Algorithm details	**Model Details** Model Name: google/mobilebert-uncased (baseline); alternatives: distilbert-base-uncased, bert-base-uncased Model Type: Transformer token-classification (binary token labels) Input Length: 64 (baseline) / 96 (larger profiles) Embedding Dimension: 384 Framework: PyTorch/Hugging Face for training → TensorFlow → TFLite for export Purpose: Detect keywords at token/subtoken resolution for on-device use **Algorithms** **Preprocessing & labeling:** normalize text → RAKE pseudo-keyword extraction → fast tokenization → align word → first-subtoken labels **Training:** fine-tune pretrained backbones; AdamW + linear warmup schedule; cross-entropy (optional inverse-frequency class weights); gradient accumulation, clipping, early stopping **Decoding**: argmax or tuned keyword-probability threshold (threshold tuned on validation for best F1) **Export & validation:** convert checkpoint → TFLite (float32 + dynamic quant); validate keyword outputs vs PyTorch on sample SMS
Is Mid review, end review report uploaded on Git ?	Yes, All mid-review and monthly connect presentations are uploaded to the repo.
Link for Git	https://github.ecodesamsung.com/SRIB-PRISM/25PI05TU_On_device_Topic_modelling_for_small_text_data

Work-let Closure Details

- **Data details:**

Items	Details
Name & Type of Data	Public SMS text corpus (short SMS messages)
Number of data points	5,574 messages (original public corpus)
Source of Data (self collected, Scrapped, available on open source)	UCI SMS Spam Collection (public). Example source: https://archive.ics.uci.edu/ml/machine-learning-databases/00228/
Google drive link/ git link to access data	Preprocessed data and scripts are available in the repository

Note: If data uploaded on google drive, access to be shared to prism.srib@gmail.com

Thank you